



Data 230 Final Project

By: Manuel Lopez

Topic: Electric Car Population Data

The screenshot shows a web browser window with multiple tabs. The active tab is 'Dataset - Catalog' with the URL 'https://catalog.data.gov/dataset?q=cars'. The page is the official website of the United States government, Data.gov. The header includes the Data.gov logo, navigation links for DATA, METRICS, OPEN GOVERNMENT, and CONTACT, and a User Guide link. The main content area is titled 'DATA CATALOG' and shows search results for 'cars'. A search bar contains the text 'cars' and a magnifying glass icon. To the right of the search bar is an 'Order by:' dropdown menu set to 'Popular'. Below the search bar, a message states: 'For a list of search operators, please see the "Search in Detail" instructions.' On the left side, there is a 'Filter by location' section with a 'Clear' link and a map of the United States. The map shows the United States with a red dot indicating the location of the search results. Below the map, it says 'OpenStreetMap contributors'. The main content area displays '801 datasets found for "cars"'. The first result is 'Electric Vehicle Population Data' with a flame icon. It is labeled 'State' and describes the dataset as showing Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) registered through the Washington State Department. Below the description are links for 'CSV', 'RDF', 'JSON', and 'XML'. The second result is 'Crash Reporting - Drivers Data' with a flame icon. It is labeled 'County' and describes the dataset as providing information on motor vehicle crashes in Montgomery County of Maryland.

Montclair Syllabus Dataset - Catalog Stats Final Project - Posit C Final Project Presentation lopezm75/Stats-230-Final

https://catalog.data.gov/dataset?q=cars

An official website of the United States government [Here's how you know](#)

DATA.GOV DATA METRICS OPEN GOVERNMENT CONTACT User Guide

DATA CATALOG / Datasets Organizations

cars

Order by: Popular

For a list of search operators, please see the "Search in Detail" instructions.

Filter by location Clear

Enter location...

United States

OpenStreetMap contributors

Topics

801 datasets found for "cars"

Electric Vehicle Population Data 🔥

State of Washington — This dataset shows the Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) that are currently registered through Washington State Department...

CSV RDF JSON XML

Crash Reporting - Drivers Data 🔥

Montgomery County of Maryland — This dataset provides information on motor vehicle

Reading the CSV Data

```
{r}  
library(tidyverse)  
library(effects)  
Evehicles <- read_csv("Electric_Vehicle_Population_Data.csv")
```

Rows: 239747 Columns: 17

— Column specification —

Delimiter: ",",

chr (11): VIN (1-10), County, City, State, Make, Model, Electri...

dbl (6): Postal Code, Model Year, Electric Range, Base MSRP, L...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.



Question 1

Is there a correlation between the Model Year and Electric Range of a car?



Question 2

Do different electric vehicle companies produce models with different electric ranges?



Descriptive Analysis

Histogram of Electric Range:

Histogram of Electric Range:

```
✓ {r}  
Evehicles |>  
  ggplot(aes(x = `Electric Range`)) +  
  geom_histogram() +  
  facet_wrap(~Make)
```



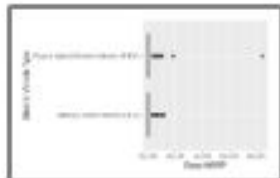
Histogram of Electric Range:



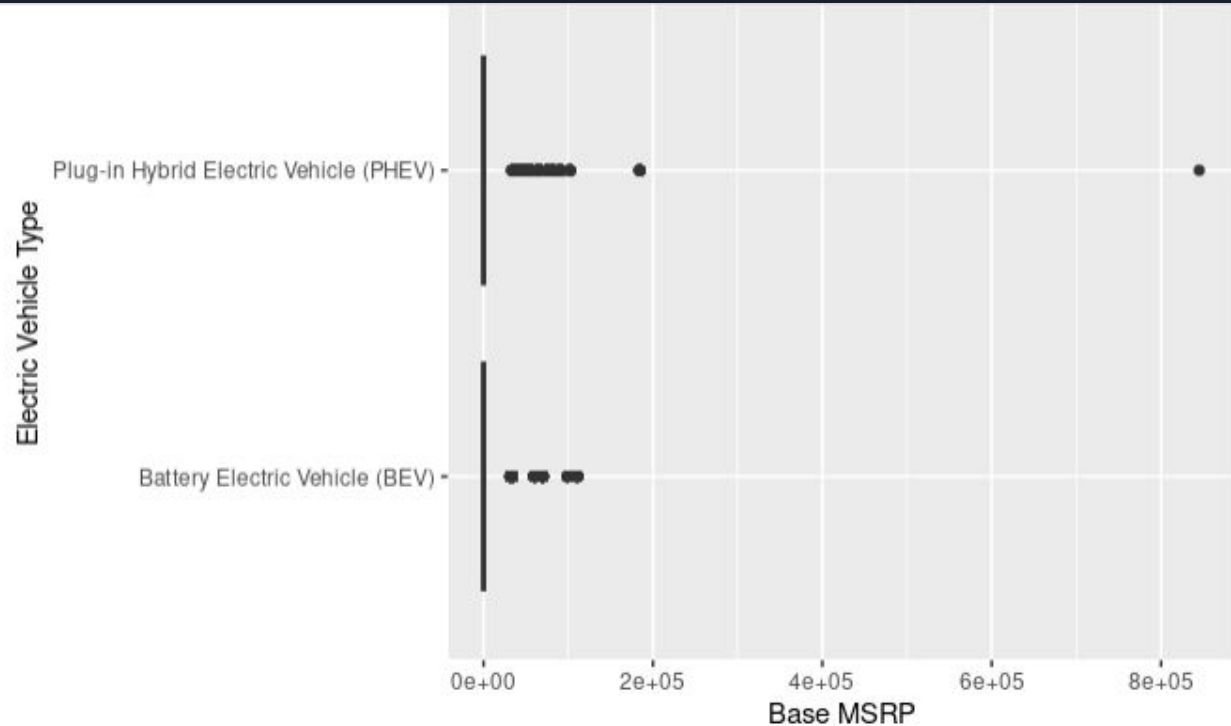
Boxplot of MSRP:

Boxplot of MSRP:

```
{r}  
Evehicles |>  
  ggplot(aes(x = `Base MSRP`, y = `Electric Vehicle Type`)) +  
  geom_boxplot()
```



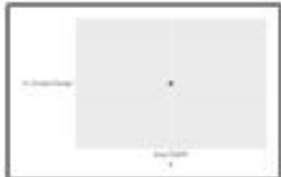
Boxplot of MSRP:



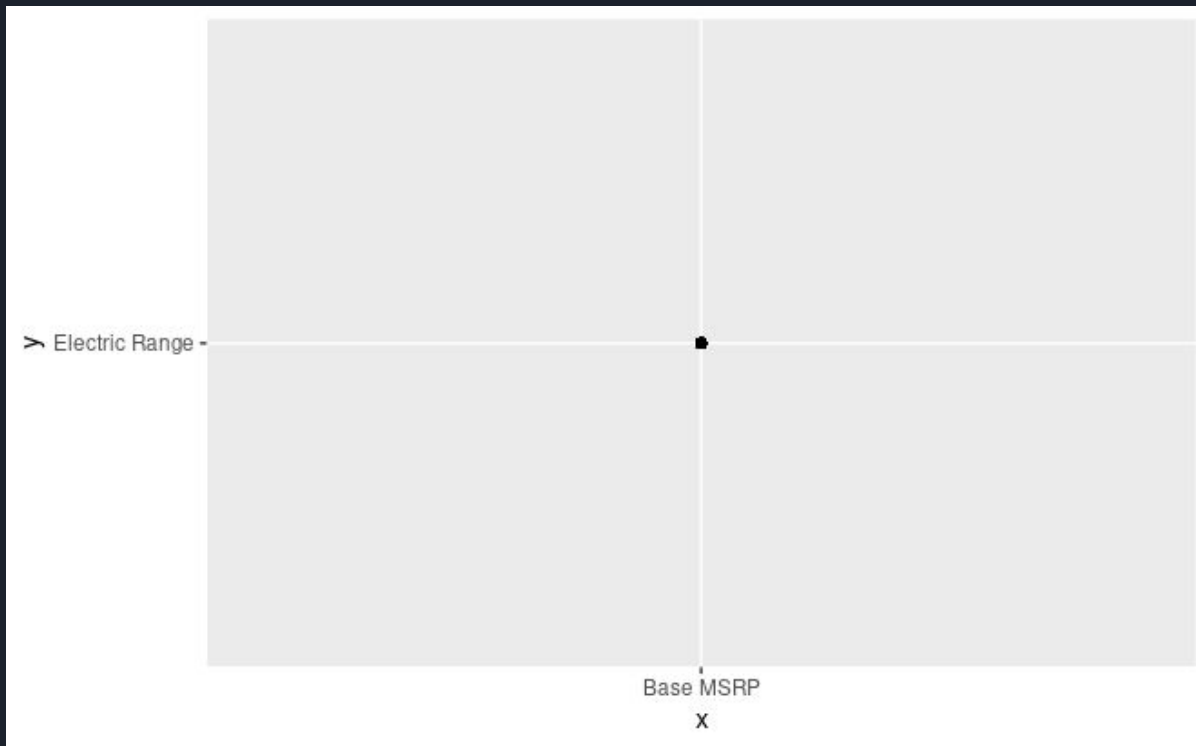
Scatterplot of Electric Range vs MSRP:

Scatterplot of Electric Range vs MSRP:

```
{r}  
Evehicles |>  
  ggplot(aes(x = "Base MSRP",  
             y = "Electric Range")) +  
  geom_point()
```



Scatterplot of Electric Range vs MSRP:





Summary

Summary Code:

Summary:

```
{r}  
Evehicles |>  
  group_by(Make) |>  
  summarize(mean = mean(`Base MSRP`, na.rm = TRUE),  
            median = median(`Base MSRP`, na.rm = TRUE),  
            min = min(`Base MSRP`, na.rm = TRUE),  
            max = max(`Base MSRP`, na.rm = TRUE),  
            sd = sd(`Base MSRP`, na.rm = TRUE))
```

Table: (1)

A tibble: 46 × 6

Make <chr>	mean <dbl>	median <dbl>	min <dbl>	max <dbl>
ACURA	0.0000	0	0	0
ALFA ROMEO	0.0000	0	0	0
AUDI	0.0000	0	0	0
AZURE DYNAM...	0.0000	0	0	0
BENTLEY	0.0000	0	0	0
BMW	2653.1263	0	0	91250
BRIGHTDROP	0.0000	0	0	0
CADILLAC	819.6767	0	0	75095
CHEVROLET	0.0000	0	0	0
CHRYSLER	1268.0053	0	0	39995

1-10 of 46 rows | 1-5 of 6 columns

Previous 1 2 3 4 5 Next

A tibble: 46 × 6

	mean <dbl>	median <dbl>	min <dbl>	max <dbl>	sd <dbl>
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	2653.1263	0	0	91250	11756.143
	0.0000	0	0	0	0.000
	819.6767	0	0	75095	7805.954
	0.0000	0	0	0	0.000
	1268.0053	0	0	39995	7008.496

1-10 of 46 rows | 2-6 of 6 columns

Previous 1 2 3 4 5 Next

Table: (2)

A tibble: 46 × 6

Make <chr>	mean <dbl>	median <dbl>	min <dbl>	max <dbl>
DODGE	0.0000	0	0	0
FIAT	0.0000	0	0	0
FISKER	6219.5122	0	0	102000
FORD	0.0000	0	0	0
GENESIS	0.0000	0	0	0
GMC	0.0000	0	0	0
HONDA	0.0000	0	0	0
HYUNDAI	0.0000	0	0	0
JAGUAR	0.0000	0	0	0
JEEP	0.0000	0	0	0

11-20 of 46 rows | 1-5 of 6 columns

Previous 1 2 3 4 5 Next

A tibble: 46 × 6

	mean <dbl>	median <dbl>	min <dbl>	max <dbl>	sd <dbl>
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	6219.5122	0	0	102000	24481.882
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000

11-20 of 46 rows | 2-6 of 6 columns

Previous 1 2 3 4 5 Next

Table: (3)

A tibble: 46 × 6

Make <chr>	mean <dbl>	median <dbl>	min <dbl>	max <dbl>
KIA	1544.1815	0	0	33950
LAMBORGHINI	0.0000	0	0	0
LAND ROVER	0.0000	0	0	0
LEXUS	0.0000	0	0	0
LINCOLN	0.0000	0	0	0
LUCID	0.0000	0	0	0
MAZDA	0.0000	0	0	0
MERCEDES-BENZ	0.0000	0	0	0
MINI	5173.5216	0	0	36900
MITSUBISHI	0.0000	0	0	0

21-30 of 46 rows | 1-5 of 6 columns

Previous 1 2 3 4 5 Next

A tibble: 46 × 6

	mean <dbl>	median <dbl>	min <dbl>	max <dbl>	sd <dbl>
	1544.1815	0	0	33950	6890.648
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	5173.5216	0	0	36900	12810.301
	0.0000	0	0	0	0.000

21-30 of 46 rows | 2-6 of 6 columns

Previous 1 2 3 4 5 Next

Table: (4)

A tibble: 46 × 6

Make <chr>	mean <dbl>	median <dbl>	min <dbl>	max <dbl>
MULLEN AUTO...	0.0000	0	0	0
NISSAN	0.0000	0	0	0
POLESTAR	0.0000	0	0	0
PORSCHE	2952.6073	0	0	845000
RAM	0.0000	0	0	0
RIVIAN	0.0000	0	0	0
ROLLS-ROYCE	0.0000	0	0	0
SMART	0.0000	0	0	0
SUBARU	1128.8710	0	0	34995
TESLA	1019.3273	0	0	110950

31-40 of 46 rows | 1-5 of 6 columns

Previous 1 2 3 4 5 Next

A tibble: 46 × 6

	mean <dbl>	median <dbl>	min <dbl>	max <dbl>	sd <dbl>
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	2952.6073	0	0	845000	27983.171
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	0.0000	0	0	0	0.000
	1128.8710	0	0	34995	6184.548
	1019.3273	0	0	110950	8441.870

31-40 of 46 rows | 2-6 of 6 columns

Previous 1 2 3 4 5 Next



Inferential Statistics

Question 1: Is there a correlation between the Model Year and Electric Range?

```
{r}  
cor.test(Evehicles$`Electric Range`,  
         Evehicles$`Model Year`)
```

Pearson's product-moment correlation

```
data: Evehicles$`Electric Range` and Evehicles$`Model Year`  
t = -307.86, df = 239696, p-value < 2.2e-16  
alternative hypothesis: true correlation is not equal to 0  
95 percent confidence interval:  
 -0.5351821 -0.5294442  
sample estimates:  
      cor  
-0.5323193
```

Question 2: Do different electric vehicle companies produce models with different electric ranges?

```
{r}
anova_result <- aov(Evehicles$`Electric Range` ~
                    Evehicles$`Make`, data = Evehicles)
summary(anova_result)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Evehicles\$Make	45	1.392e+08	3093130	484.7	<2e-16	***
Residuals	239652	1.529e+09	6382			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

49 observations deleted due to missingness



Conclusions



Conclusion for 1

Newer EV models typically have shorter electric ranges, according to the correlation analysis between Model Year and Electric Range, which showed a statistically significant negative link with a correlation coefficient of -0.5323 and a p-value $< 2.2e-16$. This trend suggests possible industry developments, such as a focus on efficiency, cost, or urban mobility, rather than maximizing range, which runs counter to the conventional wisdom that technology advancements will result in longer ranges. The narrow confidence interval (-0.5352 to -0.5294) and high absolute t-value (-307.86) further support the validity of this result. To learn more about the variables behind this pattern, future investigations could look at trends unique to a given manufacturer, developments in battery technology, or shifts in consumer desires.



Conclusion for 2

The analysis's F-value of 484.7 and incredibly low p-value ($<2e-16$) demonstrate that electric vehicle ranges differ greatly amongst manufacturers. These findings show that variations in electric range between manufacturers are statistically significant rather than random. The residual sum of squares ($1.529e+09$) explains the remaining variability within each group, although the manufacturers' sum of squares ($1.392e+08$) indicates a significant contribution to the overall variation. The difference is further emphasized by the mean square values, which show that manufacturers have an average variance of 3,093,130 while individual automobiles have an average variation of 6,382. These results imply that certain businesses give long-range EVs priority, while others concentrate on price, city driving, or other technological factors.